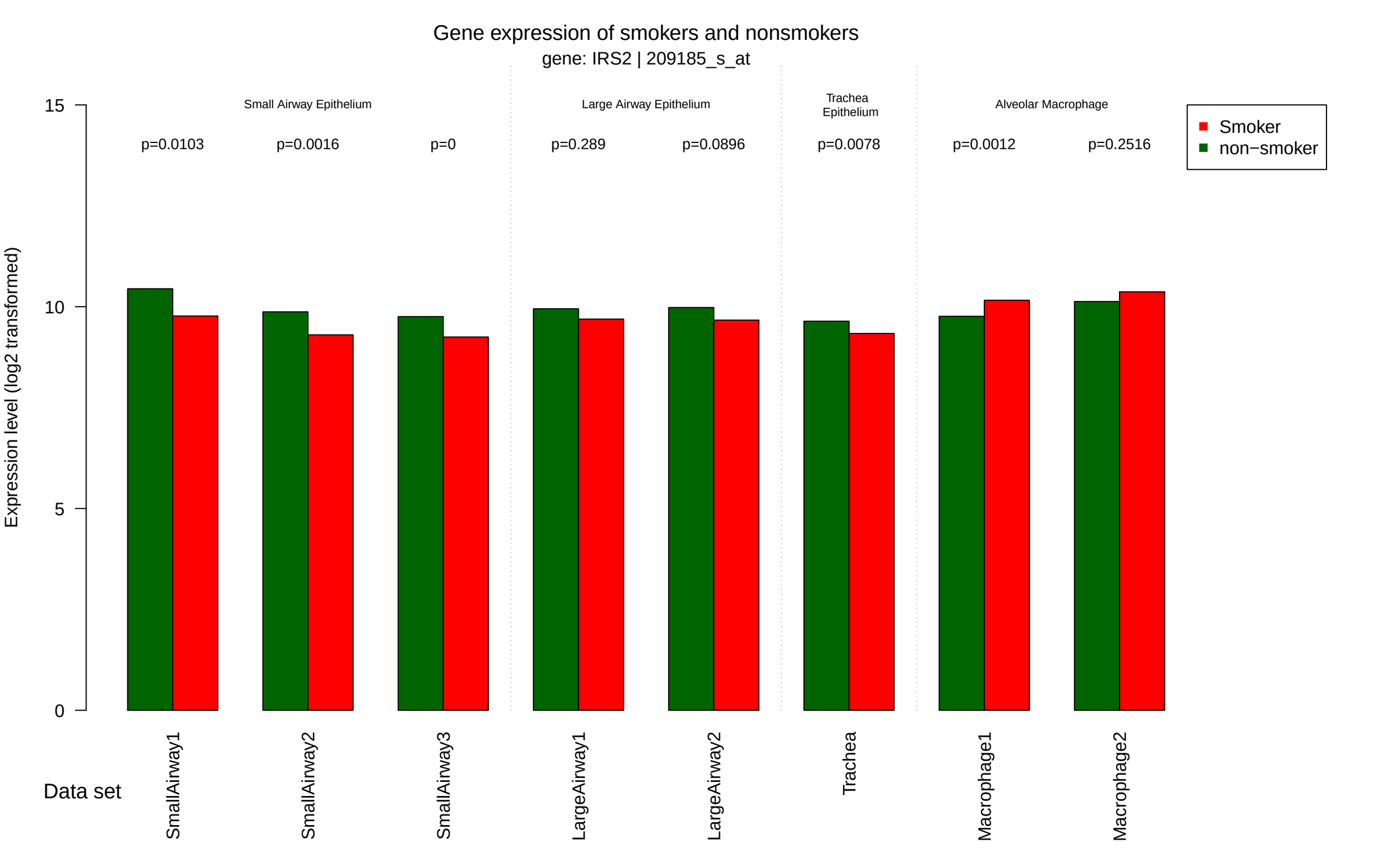
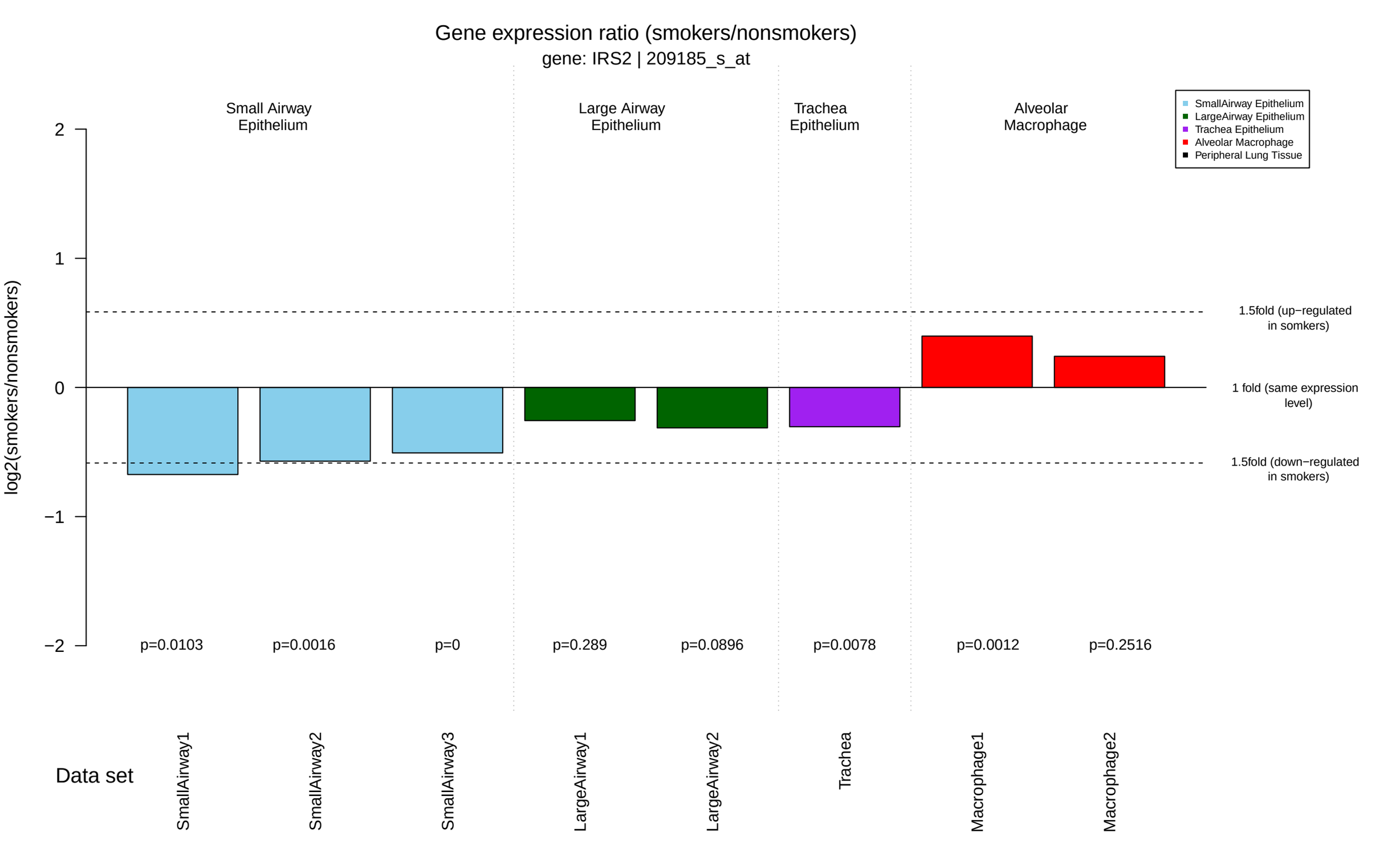
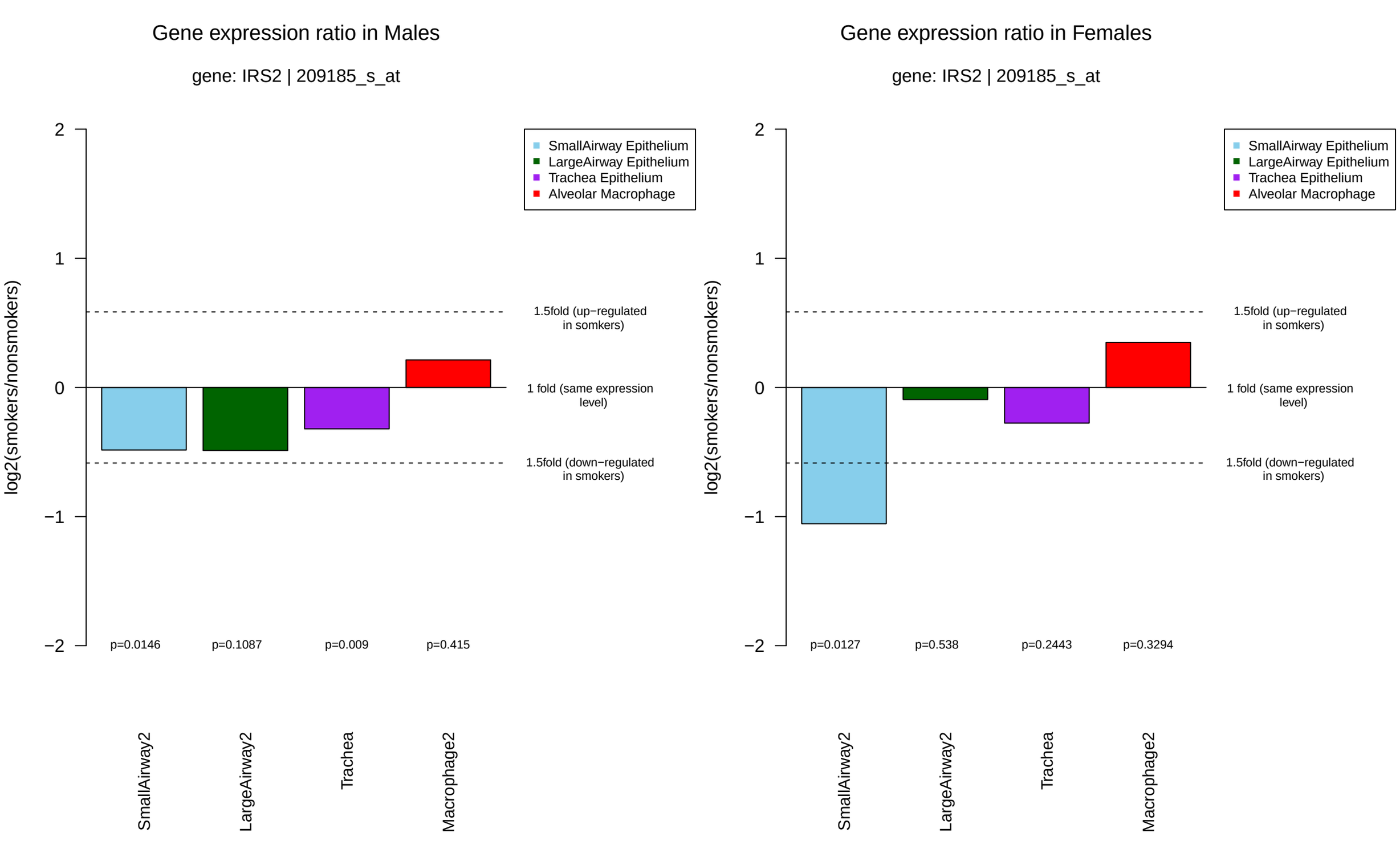
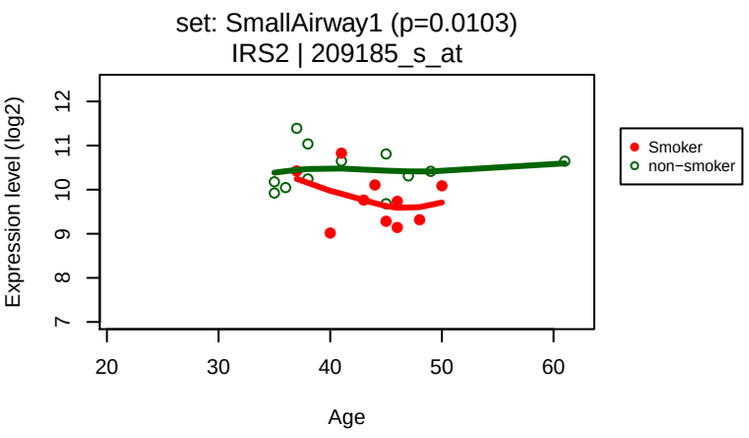
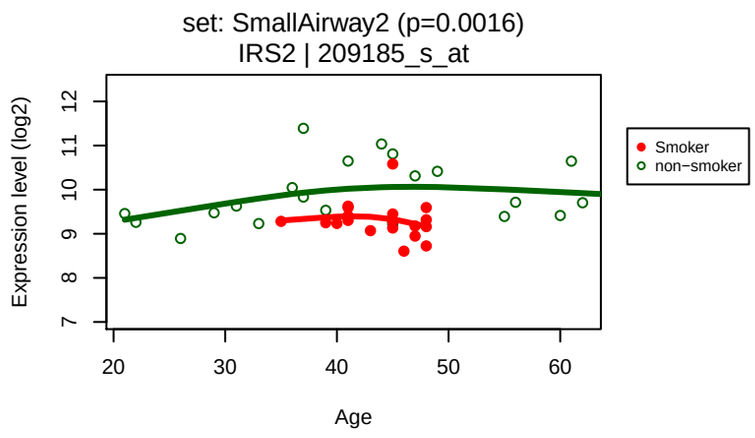
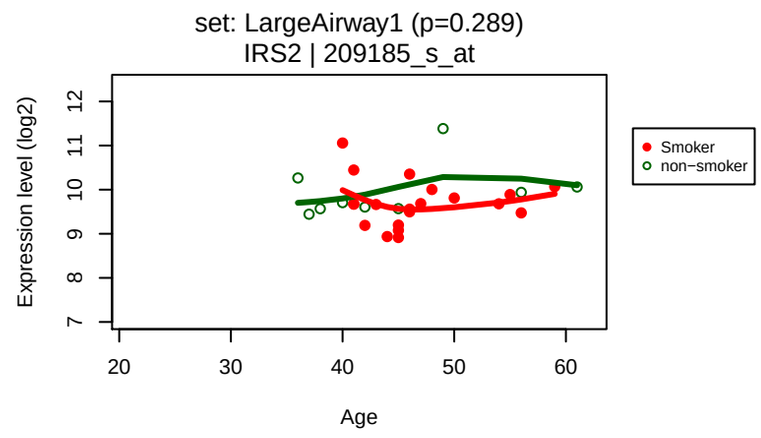

Gene expression of smokers and nonsmokers
gene: IRS2 | 209185_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0103
p=0.0016
p=0
p=0.289
p=0.0896
p=0.0078
p=0.0012
p=0.2516
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

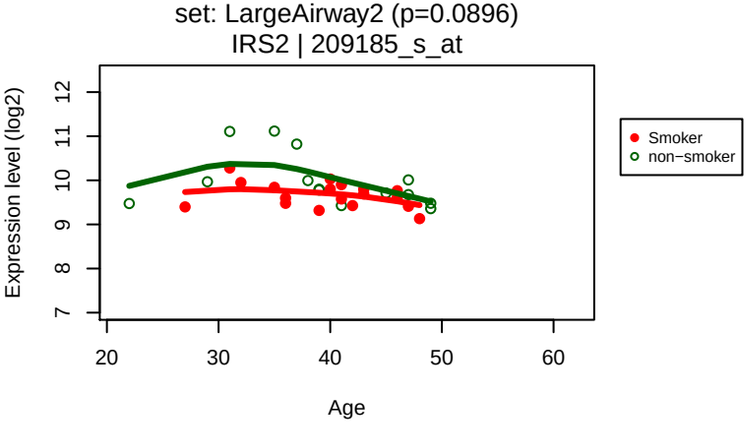

Gene expression ratio (smokers/nonsmokers)
gene: IRS2 | 209185_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0103
p=0.0016
p=0
p=0.289
p=0.0896
p=0.0078
p=0.0012
p=0.2516
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

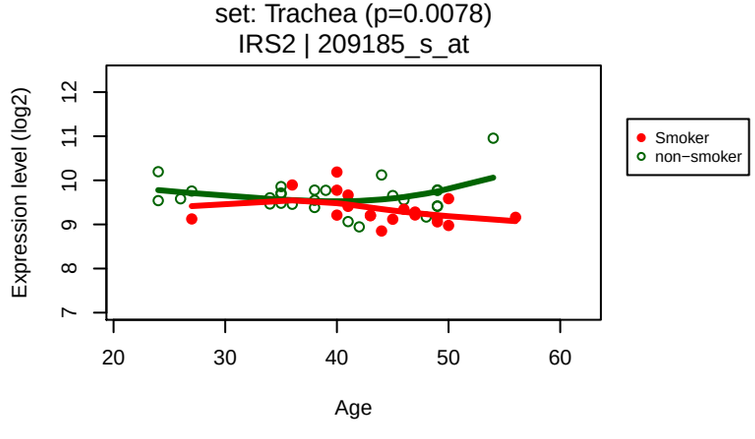

Gene expression ratio in Males
gene: IRS2 | 209185_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0146
p=0.1087
p=0.009
p=0.415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRS2 | 209185_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0127
p=0.538
p=0.2443
p=0.3294
SmallAirway2
LargeAirway2
Trachea
Macrophage2

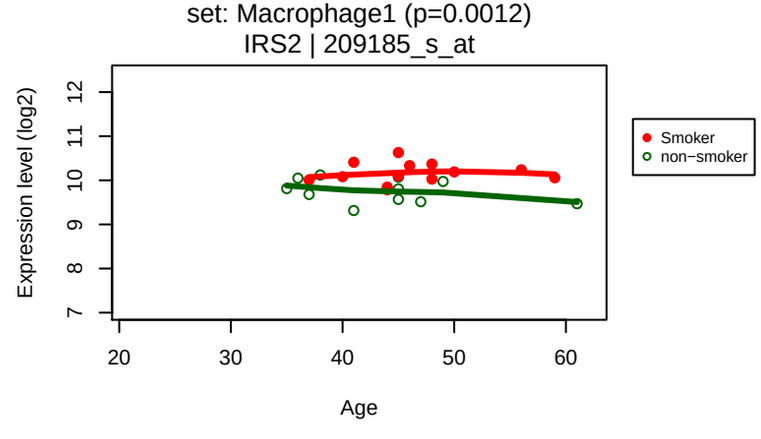

set: SmallAirway1 (p=0.0103)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker

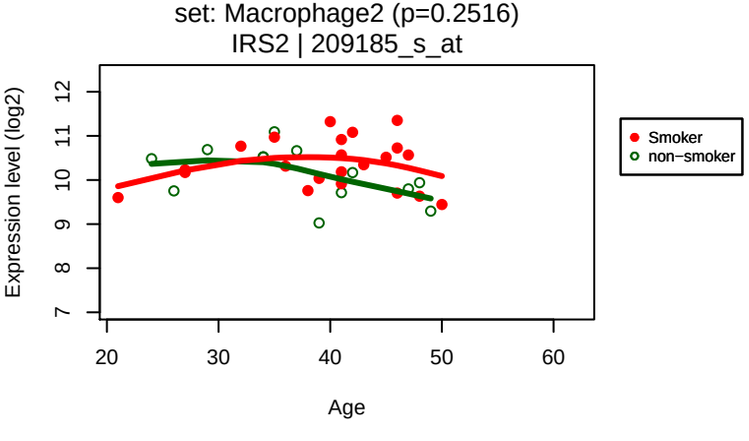

set: SmallAirway2 (p=0.0016)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker

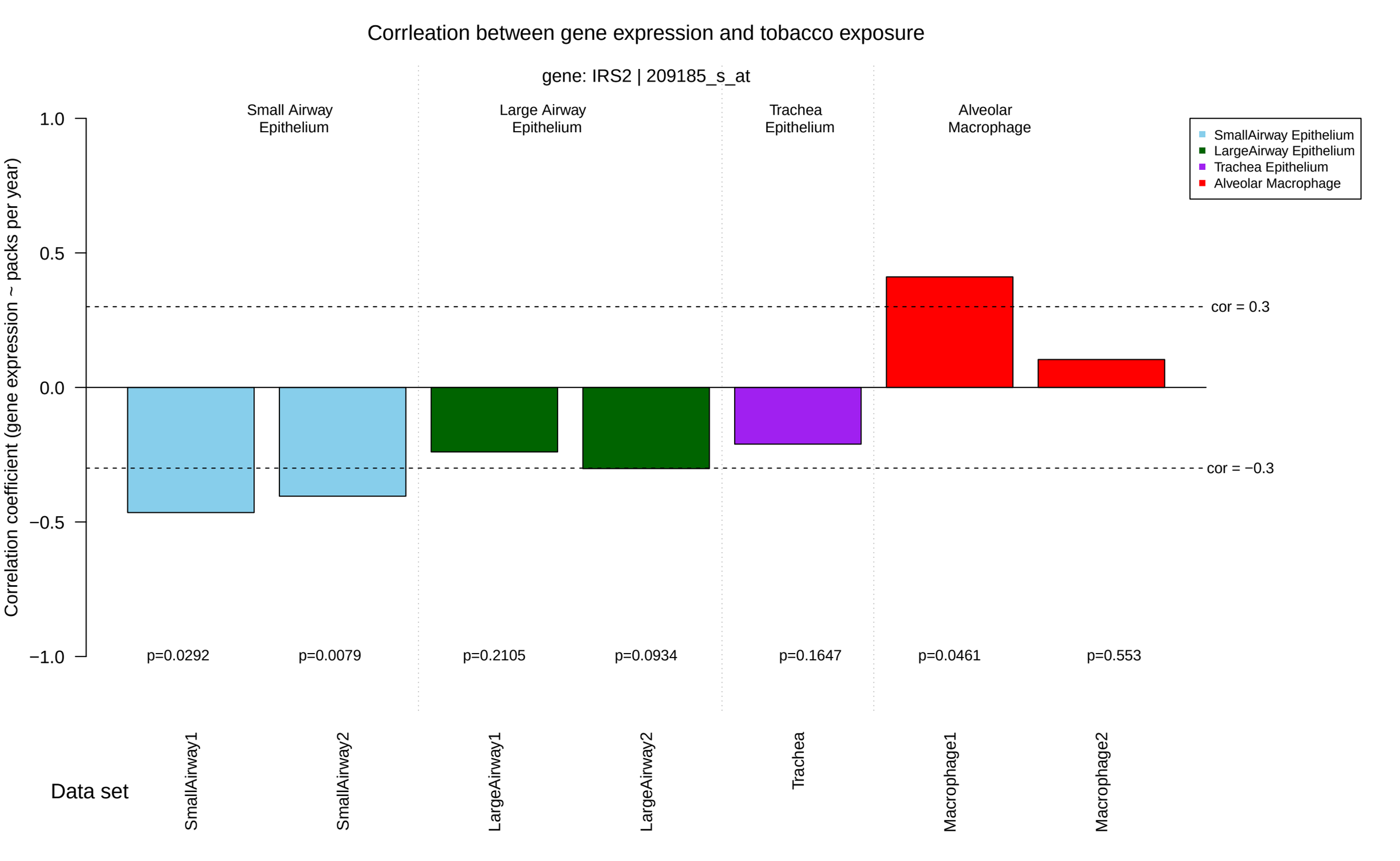

set: LargeAirway1 (p=0.289)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0896)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0078)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0012)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2516)
IRS2 | 209185_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRS2 | 209185_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0292
p=0.0079
p=0.2105
p=0.0934
p=0.1647
p=0.0461
p=0.553
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2